HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Actuarial Mathematics and Statistics

F70LB

LIFE INSURANCE MATHEMATICS B

IN-PERSON EXAMINATION

April Examination and Assessment Diet 2024

Duration: 2 hours

Total marks available: 60

SOLUTIONS

Question 1

(a)

$$a_{x|y}$$

$$a_{x|y} = a_y - a_{xy} = \left(\ddot{a}_y + 0.5\right) - \left(\ddot{a}_{xy} + 0.5\right) = 7 - 3 = 4$$

[1 mark for $\bar{a}_{x|y} = \bar{a}_y - \bar{a}_{xy}$; 1 mark correct answer]

(b)

$$q^{\,2}_{x\,y} = q_x - q^{\,1}_{x\,y} = 0.03 - 0.01 = 0.02$$

(c)

$$q_{x\,y} = q_{x\,y}{}^{1} + q^{\,1}_{x\,y} = 0.025 + 0.01 = 0.035$$

(d)

$$A_{x\,y\,:\,10\,1}{}^{1} = A_{x\,y}{}^{1} - \left(1 - q_{x\,y}\right) v^{10}\, A_{x+10:\,y+\,10}{}^{1}$$ [1 mark]

$$= 0.3 - (1 - 0.035)\, v^{10}\, 0.1 = 0.22084$$ [1 mark]

Question 2

(a) SMR = (actual deaths) / (expected deaths if standard mortality used) [1 mark]
=(1111+2441) / (111,000 * 0.0100 + 174,000 * 0.0155) = 93.3% [1 mark]

(b) The SDR will be lower than the CDR [1 mark]
… because one of the two mortality rates for age groups in the loan population is strictly lower and the other is the same. [1 mark]

(c) *(Marks given for any sensible arguments)*

(i) Temporary initial selection is a limited period of lower (or higher) decrements for lives when they first enter a population. [1 mark]
Mortality rates a significantly higher at higher durations since the beginning of the loan, and lower at shorter durations, compared to others of the same age group (and the standard population). This is evidence of temporary initial selection. [1 mark]

(ii) Class selection is a difference in decrements linked to some permanent feature of subgroups of the population, such as occupation or location. [1 mark]
Loan customers seem to have significantly different mortality to the general population. This appears to be class selection. [1 mark]

(iii) Adverse selection is where information asymmetries between the customer and the insurer lead to customers making choices that are to the disadvantage of the insurer. [1 mark]
It is not clear whether there is adverse selection. However, there is some evidence that people choosing this product have poor underlying longevity after the first few years (presumably affected by underwriting). This could lead to short loan periods and losses or lower profits. [1 mark]

(d)

(i) A semi-markov model has transition rates and probabilities which depend on the length of time in a state, not just the length of time since the start of a contract. [2 marks]

(ii) *(2 marks given for a sensible justified conclusion. 1 mark if the reasoning is not fully convincing.)* It is not necessary because transition rates can depend on time since taking out the loan without requiring the model to be semi-Markov. Our usual time-inhomogeneous Markov models are sufficient.

Question 3

(a) EPV profit = -50/1.25 + 100*(1.025^{-4} + 1.025^{-5}) = £34
[1 mark for method; 1 mark numerical answer]

(b) DPP exists, ie DPP <=5 since EPV profit>0. And DPP>3 since the first 3 years are negative or zero. So DPP is 4 or 5. [1 mark]
EPV payments up to time 4 = £34 – 100 * 1.025^{-5} = £0.96>0 so DPP = 4. [1 mark]

(c) PV premiums = 1000 + 500 * (0.8^{-2} + 0.8^{-4} + 0.8^{-6}) = £1656 [1 mark]
Profit margin = 34/1656 = 2.0% [1 mark]

Question 4

(a) ${}_tp_x^{\overline{33}} = {}_tp_x^{33} = 1$ since the lives cannot leave the dead state. [1 mark]
${}_tp_x^{\overline{11}} \neq {}_tp_x^{11}$ and ${}_tp_x^{\overline{22}} \neq {}_tp_x^{22}$ because it is possible to leave those states and return [1 mark]

(b)

$$\frac{d}{dt}({}_tp_x^{21}) = {}_tp_x^{22}\,\mu_{x+t}^{21} - {}_tp_x^{21}(\mu_{x+t}^{12} + \mu_{x+t}^{13})$$

[1 mark]

Proof:

${}_{t+dt}p_x^{21} = {}_tp_x^{21}\,{}_{dt}p_{x+t}^{11} + {}_tp_x^{22}\,{}_{dt}p_{x+t}^{21} + 0$ by law of total probability and using ${}_tp_x^{31} = 0$
[1 mark]

$= (1 - {}_{dt}p_{x+t}^{13} - {}_{dt}p_{x+t}^{12})\,{}_tp_x^{21} + {}_tp_x^{22}\,{}_{dt}p_{x+t}^{21}$ [1 mark]

$= {}_tp_x^{21}(1 - \mu_{x+t}^{13}dt + \mu_{x+t}^{12}dt + o(dt)) + {}_tp_x^{22}(\mu_{x+t}^{21}dt + o(dt))$

using Markov model assumption (ii) [1 mark]

Rearranging and dividing by dt:

$$({}_{t+dt}p_x^{21} - {}_tp_x^{21})/dt = -{}_tp_x^{21}(\mu_{x+t}^{13} + \mu_{x+t}^{12}) + {}_tp_x^{22}\,\mu_{x+t}^{21}dt + o(dt)/dt$$

[1 mark]

Taking the limit as $dt \to 0$ gives the required result since the LHS is the definition of the derivative of ${}_tp_x^{21}$ and $o(dt)/dt$ goes to zero by definition.

[1 mark]

(c)

(i) The EPV of a premium of £1000 paid whenr the life is in the Active state [1 mark]
… for the first active period only (ie, it does not resume on return to the active state) [1 mark]

(ii) The EPV of an income of £20k/year paid to the policyholder while sick [0.5 marks]
…for the first period of sickness only [0.5 marks]
…with a deferred period of 0.5 years, ie 6 months [0.5 marks]
… and paid for life (or until recovery) [0.5 marks]

(d) $50{,}000 \int_1^{15} \left({}_tp_x^{11}\, \mu_{x+t}^{13} + {}_tp_x^{12}\, \mu_{x+t}^{23} \right) e^{-\delta t} dt$
[3 marks if fully correct; 2 marks if very minor errors; 1 mark if 2 or more significant errors]

Question 5

$${}_tq\,{}^{1}_{x\,y} = \int_0^t {}_sp_x \;\; {}_sp_y\, \mu_{x+s} ds \text{ [1 mark]}$$

$$= \int_0^t e^{-10s} \;\; e^{-5s} \left(\frac{10e^{-10s}}{e^{-10s}}\right) ds \text{ [1 mark]}$$

$$= 10 \int_0^t e^{-15s} ds \text{ [1 mark]}$$

$$= 10 \left[\frac{e^{-15s}}{-15}\right]_0^t$$

$$= \frac{2}{3}(1 - e^{-15t}) \text{ [1 mark]}$$

$$= 0.5179 \text{ substituting t=0.1 [1 mark]}$$

Question 6

(a) Briefly explain what is meant by each of the following:

(i) Ill-health retirement is where a pension scheme member below the usual retirement age can retire early and receive immediate payment of their pension because they are ill. The pension may also be enhanced in some way.

(1 mark)

(ii) A member has withdrawn if they are not yet receiving their pension (eg, they are below pension age) but they are not building up any more pension. This is usually because they have left employment that is connected with the occupational pension scheme.

(1 mark)

(b) I don't agree. The reason we need to split between the different ways of exit is that different types of benefits are paid on all the different types of exit. These have different costs to provide.

(2 marks for a convincing answer; 1 mark if it is partly convincing)

Question 7

(a) *Note that since the question gives figures to 4sf, students' answers which agree to 3sf will be marked as correct.*

V(13) = 100000 $A^{@\,i=5\%}_{x+t:n-t}$ - 4000$\ddot{a}^{(4)\ @\,i=5\%}_{x+t:n-t}$ + 1000$\ddot{a}^{@\,i=1.942\%}_{x+t:n-t}$ [2 marks]

Where x=30, n=25 and t = 13.

Since the expenses increase at 3% and i=5%, we can use a net discount rate of 1.05/1.03 - 1 = 1.942% for the expenses [1 mark]

$$A^{@\,i=5\%}_{x+t:n-t} = A^{@\,i=5\%}_{x+t} + {}_{n-t}p_{x+t}\ v^{n-t}\ \left(1 - A^{@\,i=5\%}_{x+n}\right)$$

$$= 0.2346 + 1.05^{-12}\left(\frac{0.9363}{0.9868}\right)(1 - 0.3775) = 0.5635$$ [1 mark]

$$\ddot{a}^{@\,i=5\%}_{x+t:n-t} = \frac{1 - 0.5635}{d} = 9.167$$

$$\ddot{a}^{(4)\ @\,i=5\%}_{x+t:n-t} \approx \ddot{a}^{@\,i=5\%}_{x+t:n-t} - \frac{3}{8}\left(1 - {}_{n-t}p_{x+t}\ v^{n-t}\right) = 9.167 - \frac{3}{8}(1 - 1.05^{-12}\left(\frac{0.9363}{0.9868}\right))$$

= 8.990 [1 mark]

$$A^{@\,i=1.942\%}_{x+t:n-t} = A^{@\,i=1.942\%}_{x+t} + {}_{n-t}p_{x+t}\ v^{n-t}\ \left(1 - A^{@\,i=1.942\%}_{x+n}\right)$$

$$= 0.5434 + 1.01942^{-12}\left(\frac{0.9363}{0.9868}\right)(1 - 0.6631) = 0.7972$$

$$\ddot{a}^{@\,i=1.942\%}_{x+t:n-t} = \frac{1-0.7972}{d} = 10.65$$ [1 mark]

So V(13) = 100,000*0.5635 + 1030*10.65 – 4000*8.990 = 31,360 [1 mark]

$$\left(V(12.75) + \frac{4000}{4}\right)(1 + 5\%)^{0.25} = \left({}_{0.25}p_{42.75}\,V(13) + {}_{0.25}q_{42.75}\right)$$ [1 mark]

$_{0.25}p_{42.75} = \quad p_{42}^{0.25} = 0.99954$ assuming constant force of mortality [1 mark]

So

$$\mathrm{V}(12.75) = (0.99954 \times 31{,}360 \; + \; (1 - 0.99954) \times 100k)(1 + 5\%)^{-0.25} - 1000$$
$$= £30.0k$$

[1 mark]

(b) The policy value would decrease. The assumed increase in expenses is 3% which would have implied expenses of £1030 at time 13. Actual expenses of £1010 are lower so the value of the liabilities is lower. (The value of premium income is unchanged). [1 mark]

END OF SOLUTIONS